AI-assisted Project Development and Management
Seminar and Practice Group Project Guidelines

Self-check and example project development report

Based on Milestone Report requirements and content of the educational group project (using terms and conventions from the educational project guidelines)

1. Goal

The goal of the milestone report is to summarise your project development, make an initial presentation and receive comments and suggestions for further development.
The following are specific requirements:

- Report development Steps 1-5 mandatory and other steps optional
- Report on interactive work with the LLM/chatbot as design agent (details are provided below)
- Data Analytics application demo (Step 6) – Optional

2. Requirements for the Milestone Report

2.1. Recommended Structure, Content and size

- Milestone Report structure should have a main body (structured by sections), describing results of Steps 1-5, and appendices containing detailed interactive sessions with the chatbot
- It is recommended to provide in the main body of the report a summary of interaction with the chatbot, and place the detailed session record in the Appendix.
- Recommended size 15 pages, with a maximum 35 pages including the interactive session record

2.2. Report on interactive work with the LLM/chatbot as system/infrastructure design agent

The following are the expected content and topics of your report on interactive work with AI design agent:

- Prompt type selection – recommended to use a few prompt techniques (examples are provided in the practice lecture materials)
- Prompt design: structure and contextual information
- Iterative process is demonstrated by a few iterations with a chatbot/agent, with analysis of the first and following answers, instructions to correct the answer or modifying your initial/previous prompt
- Chatbot answer analysis should provide evidence of your understanding and knowledge of the course and lecture material, beneficially also additional information, reference materials or standards.
- The first chatbot answer should be provided in full; for the following iterations, it is sufficient to provide the prompt and a short analysis of the response/answer.
- The next steps in working with the chatbot may be split into some narrower topics in your project development, e.g. data ingest stage, preparation and feature engineering and so on.
- Using different chatbots and concise comparative analysis are beneficial
- The final result of your work with the design agent should normally combine a few answers and probably a line of discussion (or development/prompt tree).
- Copy&Paste of iterative session(s) is expected, but for the final report outcome it is not allowed - you need to provide a concise development summary.

3. Recommendations for AI-assisted Milestone Project Development

Follow the updated Project Development Guidelines Lecture in the Project Development folder (GoogleDrive) for the **milestone development stage**:

- Steps 1-3 are performed by the student without AI agents – to build a necessary (slide 5); they can be verified during the following Steps 4-5 with AI assistance.
- Use provided recommendations and examples of prompts and suggested prompt context, in particular slides 24-27
- You can use any chatbot you are familiar, also you can use a few chatbots to select one with which you will finalise project development, and possibly another one which may be used to verify your final development.
- It is recommended to register with your favorite chatbot and create a dedicated project, if your subscription (free or paid) allows this. This will help you with sharing the context created along the project development

The **project finalisation** Steps 6-8 are performed after the Milestone stage and may include the following activities

- Possible revision of Steps 2-3 and Steps 4-5, based on Milestone presentation, Milestone report and recommendations from the teacher
- Step 6 Demo application development – can be AI-assisted, but final code is manually verified and composed
- Steps 7-8 DMG and BDSecurity are done concisely manually following recommendations
- **Graphical materials**: Preparation of graphical materials is required to be done manually following the format and graphical icons used and recommended in Lecture and Lecture Notes materials. AI generated graphics for project technical content will be rejected with grading points penalties.

4. Reference: From the Project development guidelines - Recommended steps and sequence

Compact Steps 1-8 - Check Project Guidelines for detailed steps description.
Note: Milestone report – Development on Steps 1-5

The educational project guidelines titled “Designing the Big Data Infrastructure for your imaginary Agile Data Driven Enterprise (ADDE)”.
Goal:
The focus of the project is to implement the knowledge gained in this course, “Big Data Infrastructure Technologies for Data Analytics”, to design the Big Data Infrastructure for the imaginary Agile Data Driven Enterprise (ADDE), a new type of organisation that uses Big Data technologies to improve its business.

The project is required to implement the best practices and recommendations on building cloud based Big Data systems and Data Analytics platforms (discussed in the lectures), and also address security and compliance recommendations, including personal data protection. Implementing the agent enterprise is optional and welcome, but the main educational goal is to gain experience in structured prompt engineering with concise use of the foundational architecture design principles.

Step 1 - ADDE. Describe your imaginary company that you want to transform to the ADDE type (you can also use your employment company as an example). Select the industry sector and your particular business activity.
Step 2 - WebPresence. The company should have a web presence and you need to decide what services you provide on the web, what data you use and what data you can collect from the web. Remember that one of data types or sources used by ADDE is social media data. Provide suggestions how you can use social media data for your company.

Step 3 - DataFlow. Describe **data flows** in your company. Identify where you will use data collected from the web and other sources of information like sell statistics, user activity, or soc

Step 4 - DataAnalytics. Suggest what **data analytics methods** you will use. Search for similar use cases and solutions they use. You need to develop and demonstrate the prototype of the working application.

Step 5 - BDplatform. Select a suitable **platform** for general data management and a Big Data Analytics platformfor that can implement the data flow and data analytics processes.

Step 6 – DevDemo: Development and implementation. This step is focused on the implementation of data analysis method and preparation of a demonstration of a working algorithm. Full deployment of the algorithm on a cloud platform is not required.

Step 7 - DMG. Define a **data management policy**, including data quality, organisational data governance architecture, and procedures.

Step 8 - BDSecurity. Address **data protection and access control**, and **compliance** issues.